Resolution: Resolvent

Video Lesson

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Slide Deck

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Lesson Transcript

In this lecture, we will see how we can resolve two clauses based on all of the knowledge that we learned so far, including substitution and unification.

This is the last concept that we need to learn before we can perform a full-blown resolution-based theorem proving in first-order logic.

Resolution in Predicate Calculus

- Factors
- Binary resolvent
- Properties of resolution

We will look at the concept of factors, binary resolvent, and properties of resolution in this lecture.

Factor of a Clause

Definition: If two or more literals of a clause C (with the **same** sign) have a most general unifier σ, then $C\sigma$ is called a **Factor** of C. If $C\sigma$ is a unit clause, it is called a **Unit Factor** of C.
Example: $C = P(x) \vee P(f(y)) \vee \neg Q(x)$.

- The first two literals have a unifier $\sigma = \{x/f(y)\}$, so C has a factor $C\sigma = P(f(y)) \vee \neg Q(f(y))$.

Note: Factors of a clause are much succinct and when two clauses C_1 and C_2 cannot be resolved directly, their factors (let's call them C_1' and C_2' **can be** resolved.

Unlike in propositional logic, where a particular atom can only appear once in a clause, in first-order logic, the same predicate can appear more than once with different arguments.

This makes the resolution step a little bit more complicated because resolution calls for eliminating exactly one pair of complementary predicates.

To deal with this kind of issue, we can construct a factor from the clause to reduce multiple predicates into one.

Here is the definition of the factor of a clause. If two or more literals of a clause, *C*, with the same sign, have a most general unifier σ, then the σ applied to this clause, *C*, is called a Factor of *C*.

If C_σ is a unit clause, it is called a unit factor of *C*.

Let us consider this clause $P(x) \vee P(f(y)) \vee \neg Q(x)$. So, this predicate *P* is repeated twice, but they have different arguments. But we can see that we can find a unifier for these two predicates, which is $\sigma = \{x / f(y)\}$.

So, when we apply this substitution for the unifier to this clause, we end up with $P(f(y)) \vee P(f(y))$, which are identical, so that reduces to a single predicate. And we have this carryover, $\neg Q(x)$, where x has to be replaced by $f(y)$. So, in the end, we get $C_\sigma = P(f(y)) \vee \neg Q(f(y))$.

Factors of a clause are much more succinct, and when two clauses, C_1 and C_2, cannot be resolved directly, their factors may be able to be resolved.

Resolving Two Clauses

Definition: Let C_1 and C_2 be two clauses (called *parent clauses*) with no variables in common, and with complementary literals L_1 and L_2 such that L_1 and $\neg L_2$ have a most general unifier σ. Then the clause

$(C_1\sigma - L_1\sigma) \cup (C_2\sigma - L_2\sigma)$
is called a **binary resolvent** of C_1 and C_2. The literals L_1 and L_2 are called **the literals resolved upon**.
Note: A clause can be treated as a set of literals.
$\{P(x)\} \cup \{Q(x)\} = \{P(x), Q(x)\} = P(x) \vee Q(x)$
Example: Resolve the following (hint: $\sigma = \{x/a\}$)
$C_1 = P(x) \vee Q(x)$ and $C_2 = \neg P(a) \vee R(y)$.

Resolving two clauses can be described using this kind of set notation. Consider $P(x) \vee Q(x)$. Since the connectives are always disjunction, we can just collect these predicates, $P(x)$, $Q(x)$, and so on, into a set. So, if I have a set $\{P(x), Q(x)\}$, then that fully describes this clause. Furthermore, when we have one clause with $P(x)$ in it and another clause with $P(x)$ in it, then we can find the set union of these to get $P(x) \vee Q(x)$ for the set form for this expression.

Likewise, we can also use the set difference as shown above, which is how the resolution of two clauses is defined using this set notation. Let C_1 and C_2 be two clauses (these are called the parent clauses) with no variables in common. If we have any common variable *x* repeated here, we can simply rename it with another variable that does not occur in C_1.

And here, if we have complementary literals L_1 and L_2 such that L_1 and not L_2 have a common most general unifier σ, then use this set operation. So, we apply σ, the most general unifier, to C_1, and we apply that same unifier to C_2. So, these each would result in a set like this, and we find the set difference between that original set and a subset that includes only the complementary literal applied with the unifier.

So, this effectively removes the affected literal from Clause 1, and this effectively removes the affected literal from Clause 2. Then when we find the union, then it will collect all of the remaining predicates or literals in Clause 1 and Clause 2, after removing the complementary literal.

We can look at this example when Clause 1 is $P(x) \vee Q(x)$ and Clause 2 is $\neg P(a) \vee R(y)$, where a is a constant.

Resolvent

Definition: A *resolvent* of parent clauses C_1 and C_2 is one of the following binary resolvents:

1. a binary resolvent of C_1 and C_2
2. a binary resolvent of C_1 and a factor of C_2
3. a binary resolvent of a factor of C_1 and C_2
4. a binary resolvent of a factor of C_1 and a factor of C_2

Example: resolve the two clauses

1. $C_1 = P(x) \vee P(f(y)) \vee R(g(y))$ and
2. $C_2 = \neg P(f(g(a))) \vee Q(b)$.

(hint: resolve the factor of C_1 and clause C_2)

A resolvent of parent clauses C_1 and C_2 is one of the following binary resolvents. A binary resolvent of C_1 and C_2 directly if they are resolvable without any further modification or a binary resolvent of C_1 and a factor of C_2 or a binary resolvent of a factor of C_1 and C_2. And finally, a binary resolvent of a factor of C_1 and a factor of C_2.

Here we can consider the case where C_1 and C_2 are shown as $P(x) \vee P(f(y)) \vee R(g(y))$ for C_1 and $\neg P(f(g(a))) \vee Q(b)$ for C_2. Here we can see that we can find the factor of C_1 by combining these two into a single predicate by changing this to $f(y)$, and then we can see whether we can resolve these two.

So, this would fall under this category, a binary resolvent of a factor of C_1 and C_2.

Property of Resolution for First-Order Logic

- **Complete**: If a set of clauses S is unsatisfiable, resolution will *eventually* derive **False**.
  - *Everything that is true can be proved (eventually).*
- **Sound**: If **False** is derived by resolution, then the original set of clauses S is unsatisfiable.
  - *Everything that is proved is true.*

The full resolution-based proof for first-order logic is both complete and sound. Complete means if a set of clauses *S* is unsatisfiable, the resolution will eventually derive **False**. So, this means everything that is **True** can be proved eventually, and it is also sound. So, if **False** is derived by resolution, then the original set of clauses *S* is unsatisfiable, and in other words, this means everything that is proved by resolution is **True**.

Weakness of Resolution

Basically, resolution tries to derive

$$\textit{Axioms} \land \neg\, \textit{Theorem} = \textbf{False}$$

- Is there a **False** in the axioms? If there is, the whole formula will always be unsatisfiable no matter what.
- Can we tell whether axioms alone can derive **False** ? (generally, this is not the case)

Of course, the weakness of resolution for first-order logic is the same as that of propositional logic. Basically, resolution tries to derive axioms and the negation of the theorem for the conclusion to be **False**. But if **False** can be derived solely based on the axioms, then this would be **False**, and we can derive any conclusion from that.

Resolving Two Clauses: Revisited

Resolving two clauses C_1 and C_2 with the most general unifier σ:
$(C_1\sigma - L_1\sigma) \cup (C_2\sigma - L_2\sigma)$
This is basically:

1. Find the most general unifier σ.
2. Apply σ to both C_1 and C_2
3. Remove the complimentary literal from C_1 and C_2.

With that, let us look at the example that we saw earlier. So again, to repeat, resolving two clauses C_1 and C_2 with the most general unifier σ for the complementary literal becomes $C_{1\sigma}$. The set describes that clause and removes the effective literal and union of that with clause 2σ, where the complementary literal is again removed.

Basically, we want to find the most unifier sigma and apply sigma to both C_1 and C_2. Then we remove the complimentary literal from C_1 and C_2.

Resolving Two Clauses: Example Revisited

Example: Resolve the following (hint: $\sigma = \{x/a\}$)

$C_1 = \underbrace{P(x)}_{L_1} \vee Q(x)$ and $C_2 = \underbrace{\neg P(a)}_{L_2} \vee R(y)$

$$\begin{array}{cc}
C_1 = P(x) \vee Q(x) & C_2 = \neg P(a) \vee R(y) \\
| & | \\
\sigma = \{x/a\} & \sigma = \{x/a\} \\
\downarrow & \downarrow \\
C_1\sigma = P(a) \vee Q(a) & C_2\sigma = \neg P(a) \vee R(y) \\
| & | \\
\text{remove } L_1\sigma = P(a) & \text{remove } L_2\sigma = \neg P(a) \\
\downarrow & \downarrow \\
(C_1\sigma - L_1\sigma) = Q(a) & (C_2\sigma - L_2\sigma) = R(y)
\end{array}$$

Let us look at the example that we saw earlier. So, $P(x) \vee Q(x)$ is Clause 1, and $\neg P(a) \vee R(y)$ is Clause 2. First, we want to identify the complementary literal.

Here is $P(x) \wedge \neg P(a)$, which are not exactly the same. So, we need to find the unifier in this case, which turns out to be $\{x / a\}$. So, we apply $\{x / a\}$ to $P(x) \wedge \neg P(a)$. Then we get $C_{1\sigma}$, which is $P(a) \vee Q(a)$. And $C_{2\sigma}$ now becomes $\neg P(a) \vee R(y)$, because there was no x in this clause, y is unaffected.

Then we want to remove $L_{1\sigma}$. L_1 was this $P(x)$. So, when we apply $\{x / a\}$, then $L_{1\sigma}$ becomes $P(a)$, and here we want to remove $L_{2\sigma}$ because L_2 was not $P(a)$, and when we apply σ to it, it gives us $\neg P(a)$. So, we remove $\neg P(a)$ from $C_{2\sigma}$ and remove $P(a)$ from $C_{1\sigma}$.

Then we get $Q(a)$ on the left-hand side, and we get $R(y)$ on the right-hand side.

Finally, when we find the union of these two, again, supposing this is a set, that is a set, then we get $Q(a) \vee R(y)$. To summarize, $\{Q(a)\} \cup \{R(y)\}$ becomes a set containing $Q(a)$ and $R(y)$, and that is, by definition, $Q(a) \vee R(y)$.

In this example, we did not need to find any factor because there was exactly one pair and no repeated predicates within each clause.

Resolving Two Clauses: Example Revisited

Example: Resolve the following (hint: $\sigma = \{x/a\}$)

$C_1 = \underbrace{P(x)}_{L_1} \vee Q(x)$ and $C_2 = \underbrace{\neg P(a)}_{L_2} \vee R(y)$

$$
\begin{array}{cc}
C_1 = P(x) \vee Q(x) & C_2 = \neg P(a) \vee R(y) \\
\mid & \mid \\
\sigma = \{x/a\} & \sigma = \{x/a\} \\
\downarrow & \downarrow \\
C_1\sigma = P(a) \vee Q(a) & C_2\sigma = \neg P(a) \vee R(y) \\
\mid & \mid \\
\text{remove } L_1\sigma = P(a) & \text{remove } L_2\sigma = \neg P(a) \\
\downarrow & \downarrow \\
(C_1\sigma - L_1\sigma) = Q(a) & (C_2\sigma - L_2\sigma) = R(y)
\end{array}
$$

Now let us consider an example where we have to find the factor. This is the same example that we saw earlier. So, Clause 1 is $P(x)$ or $P(f(y))$ or $R(g(y))$, and Clause 2 is not $P(f(g(a)))$ or $Q(b)$. Since Clause 1 is the only one where the same predicate is repeated, we want to find the factor of it, and we quickly see that $\{x / f(y)\}$ is a unifier for these two.

So, we apply this to Clause 1 to get $P(f(y))$ or $R(g(y))$.

Now, we have the factor of Clause 1 here and Clause 2 by itself. And we find that there are P and ¬P, and we want to resolve them, but again, they are not directly resolvable because there is a disagreement between $\{ y / g(a) \}$. So, we use that to construct the unifier and apply the unifier to both clauses, the entire thing, this and that one.

So here, y and y. These are replaced by $g(a)$, and then here, there is no y, so the same clause is carried over. Then finally, when we remove these two complementary literals that are unified, then we can retrieve $R(g(g(a)))$ from Clause 1 and $Q(b)$ from Clause 2 to construct the final resolvent.

With the materials that we have learned so far, we are now finally ready to perform a full-blown resolution-based theorem-proving example.

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